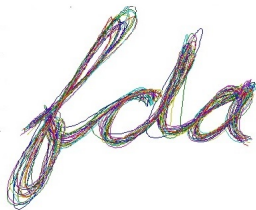


Functional data analysis

1. Introduction, motivating examples

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Lecture notes, 2026



Outline of the lecture

① *Functional data*

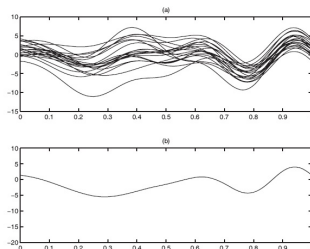
② *Examples*

Objectives:

- ▶ *to provide an understanding of how functional data arise and the areas where it is commonly encountered,*
- ▶ *to present motivating examples.*

Functional data

- ▶ Functional data analysis (FDA) deals with the analysis and theory of data in the form of functions, images and shapes, or more general objects.
- ▶ The atom of functional data is a function, where for each subject in a random sample one or several functions are recorded.



These curves were simulated without adding measurement errors, and they look smooth and continuous.

- ▶ Functional data analysis is, from our perspective, the statistical analysis of sample path data observed from continuous time stochastic processes.
- ▶ Thus, we are dealing with random processes whose realizations fall into some suitable collection of functions.

Definition [Functional Data]

A sample

$$x_1, \dots, x_n$$

is called *functional data sample* when the i 'th observation is a real-valued function

$$x_i(t), \quad t \in \mathcal{T} \subset \mathbb{R},$$

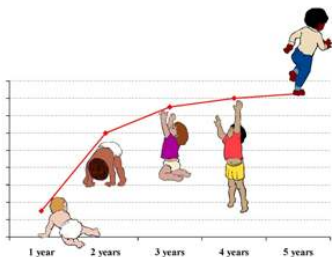
and hence, each x_i is a point in some function space.

- ▶ A well accepted *theoretical model* for this type of data is to consider it as sample paths of stochastic processes

$$X_1, \dots, X_n, \quad X_i = (X_i(t), t \in \mathcal{T}).$$

- ▶ With the advance of modern technology, more and more data are being recorded continuously during a time interval or intermittently at several discrete time points.
- ▶ They are both examples of "functional data", which have become a prevailing type of data.
- ▶ Functional Data Analysis (FDA) encompasses the statistical methodology for such data.
- ▶ Broadly interpreted, the FDA deals with the analysis and theory of data in the form of functions.

Example 1: Human Growth Data I



- ▶ The careful documentation of human growth is essential to define what we call normal growth so that we can detect as early as possible when something is going wrong with the growth process.
- ▶ Auxologists, the scientists specialising in the study of growth, also need high-quality data to advance our understanding of how the body regulates its growth.
- ▶ It may come as a surprise to learn that human growth at the macro level we see in our children is poorly understood.

Example 1: Human Growth Data II

- ▶ Growth data over the entire growing period are exceedingly expensive to collect since children must be brought into the laboratory at preassigned ages over about a 20-year span. Meeting this observational regime requires great dedication and persistence by the parents. The dropout rate is understandably high, even taking for granted the long-term commitment of maintaining a growth laboratory.
- ▶ The Fels Institute in Ohio, for example, has been collecting growth data since 1929 and is now measuring the third generation for some of its original cases.
- ▶ The accurate height measurement is also difficult and requires considerable training. Height diminishes throughout the day as the spine compresses, but it also depends on other factors. Infants must be measured lying down, and when the transition is made to measuring their standing height, measurements shrink by around one centimetre. The most careful procedures

Example 1: Human Growth Data III

still exhibit standard deviations over repeated measurements of about three millimetres.

- ▶ Records of a child's height over 20 years display features that are difficult for a data analyst to model. The classic approach has been to use mathematical functions depending on a limited number of unknown constants, and auxologists have shown much ingenuity in developing parametric models to capture these features. The best models have eight or more parameters and are still viewed as possibly missing some aspects of actual growth.
- ▶ Non-parametric modelling techniques developed over the last three decades, such as kernel and spline smoothing methods, have been applied to growth data.

Example 1: Human Growth Data IV

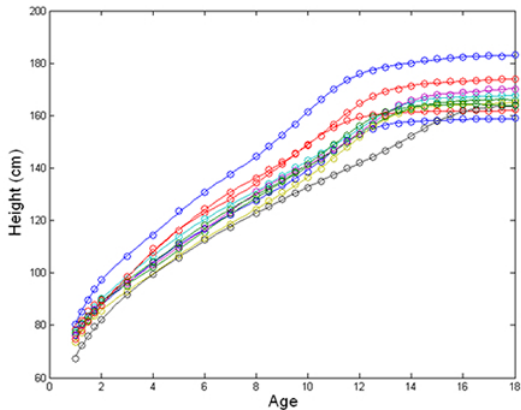
- ▶ These methods have been successful at detecting new features missed by parametric models, but they are not guaranteed to produce smoothing curves that are monotonic or strictly increasing.
- ▶ Even a small failure of monotonicity in a height curve can have serious consequences for the corresponding growth velocity and even more so for acceleration curves, which are especially important in identifying processes regulating growth.

Example 1: Human Growth Data V

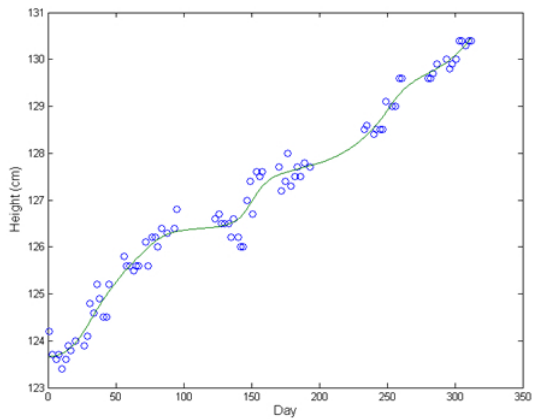
- ▶ Figure below shows, for each of 10 girls, the height function $H(t)$ as estimated from 31 observations taken between 1 and 18 years.

Note that the times of measurement are not equally spaced. Growth is the most rapid in the earliest years, but we note the increase in slope during the pubertal growth spurt (PGS) that occurs at ages ranging from about 9 to 15 years. One girl is tall for all ages, but some girls can be tall during childhood but have a comparatively small adult stature. The intervals between height measurements are six months or more, and the picture from this long-term perspective is of a relatively smooth growth process.

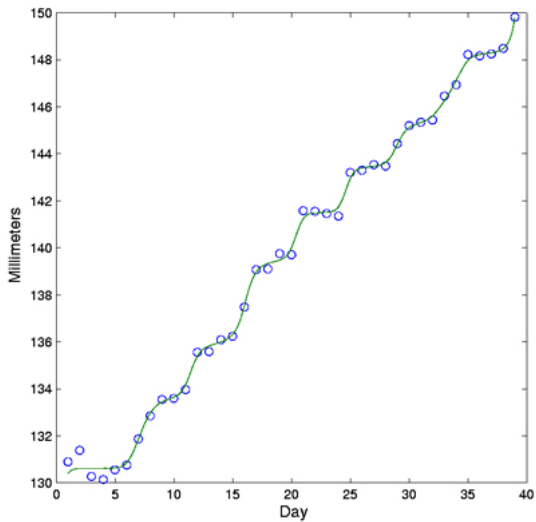
Example 1: Human Growth Data VI



- ▶ The figure below zooms in on growth by using measurements of a boy's height at 83 days over one school year, with gaps corresponding to the school vacations.
- ▶ The measurement noise in the data, with a standard deviation of about 3 mm, is apparent. The trend is also noticeably bumpier, with height increasing more rapidly over some weeks than others.

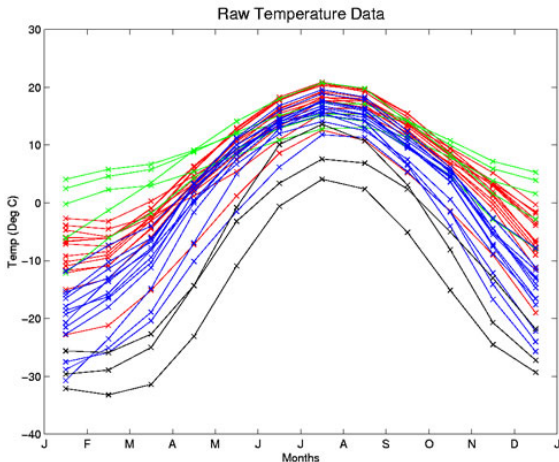


- ▶ To zoom in further, more accurate measurements are essential. The length of the tibia of a baby measured to within about 0.1 mm is graphed in Figure below.
- ▶ The jumps or saltations we saw in the boy's growth are now much more visible. These data demand that we find a way to estimate just how much bone length changes over, say, a 24-hour period. Since bone length can only increase, it is essential that any smooth line, such as that in the figure, also be everywhere increasing, and this is one of the features of the smoothing method we use.



- ▶ In these growth examples, we look at new growth data analysis developments.
- ▶ A developed method for monotonic smoothing is applied to some old and new data.
- ▶ Another aspect of the analysis is the introduction of curve registration methods, which allow the separation of amplitude and phase variation.

Example 2: Weather Data Analysis



Raw data for mean monthly temperatures at thirty-five Canadian weather stations. Data for Atlantic stations are plotted in red, Continental in blue, Pacific in green, and Arctic in black.

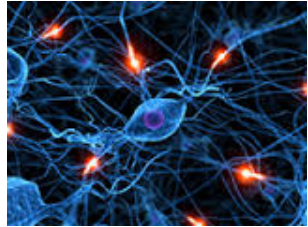
Climatologists may ask:

- ▶ What's the typical weather pattern for a Canadian city? For a city on the Pacific coast? In the Arctic region?
- ▶ What months have the most variable weather across Canada? The most similar weather?
- ▶ Are the summer temperatures just a mirror image of the winter temperatures? That is, are the weather patterns balanced in the length of seasons and in their intensity? Or do some cities tend to have extreme temperatures that last for a long time?
- ▶ How do the shapes of the weather patterns differ among the Pacific, Continental, Atlantic, and Arctic climates? How does the weather pattern in Montreal differ from the typical pattern in the rest of the Atlantic cities?

Statisticians, on the other hand, will have different questions to ask of the data. For example:

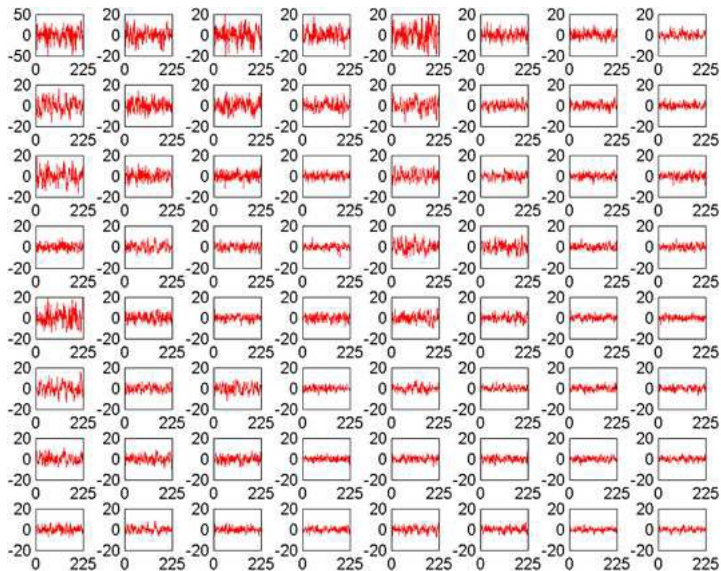
- ▶ How can we represent the temperature pattern of a Canadian city over the entire year instead of just looking at the twelve discrete points? Should we "connect the dots," or is there a better way to do this?
- ▶ Do the summary statistics "mean" and "covariance" have any meaning when dealing with curves?
- ▶ How can we determine the primary modes of variation in the data?
- ▶ How many typical modes can summarize these thirty-five curves?
- ▶ Do these curves exhibit strictly sinusoidal behaviour?
- ▶ Can we perform an analysis of variance (ANOVA) or linear model with the curves as the response and the climate as the main effect?

Example 3: Functional Data Analysis in Brain Imaging Studies



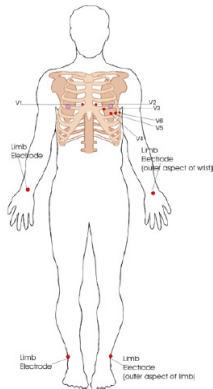
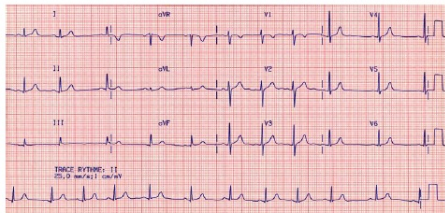
- Among many of the most popular FDA applications, functional neuroimaging has created revolutionary movements in medicine and neuroscience.

- ▶ Modern functional brain imaging techniques, such as PET (positron emission tomography), fMRI (functional magnetic resonance imaging), EEG (electro-encephalography) and MEG (magneto-encephalography), have been used for assessing the functional organization of the human/animal brain.
- ▶ These techniques allow real-time observation of the underlying brain processes during different experimental conditions, and have emerged as a crucial tool for understanding and assessing new therapies for neurological disorders.
- ▶ These measurements, called time courses, can be treated as functions of time.



Functional Data Analysis of ECG signals: biostatistical and numerical applications I

The main interest is in the statistical analysis of human ECG signals, with emphasis on the prediction of possible pathological states.



Functional Data Analysis of ECG signals: biostatistical and numerical applications II

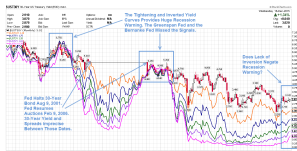
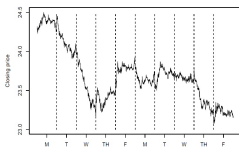
The ECG is the record in time of the electric the activity of the heart:

- ▶ is collected via electrodes placed on different points of the body,
- ▶ is a simple and non-intrusive way to assess the overall cardiac health of a subject,
- ▶ is composed of 12 leads.

Example 4: Financial Econometrics I

- In the recent **econometric literature** there has been an increasing interest in models for functional variables.
- Typical examples in finance are
implied volatility functions, **yield curves**, risk-neutral densities, **stock prices**, **returns of stock indexes**, daily log returns, **cumulative intraday returns** ect.;

Microsoft stock prices in one-minute resolution

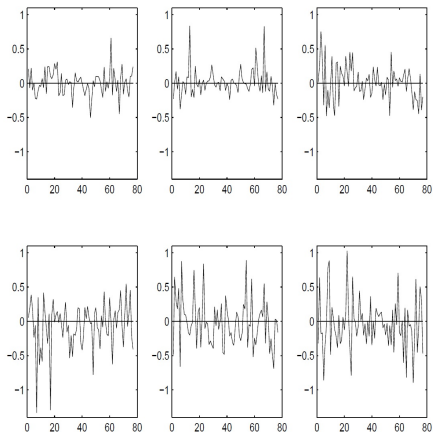


Ultra high-frequency data: shortly Recently, in the context of financial markets, there has been an increasing availability of intra-daily and ultimately transaction data called (by Engle (2000)) ultra-high-frequency (UHF) data. Modelling this kind of data may have important implications for market microstructure theory.

Financial returns I

- ▶ Probably the first paper proposing to use FDA in the context of financial returns was Anderson and Newbold (2002) who model the time-varying unconditional distribution of intra-daily returns of the Swiss-Franc-Dollar exchange rates.
- ▶ Müller, Stadtmüller and Yao (2006) and Müller, Sen and Stadtmüller (2006) propose FDA estimate and predict the volatility during the second part of the day using information contained in the first part of the day, respectively.
- ▶ Alva, Romo and Ruiz (2009) proposed functional data analysis techniques to study the intra-daily volatility. In particular, the volatility extraction is based on functional principal components and the volatility prediction on functional AR(1) models.

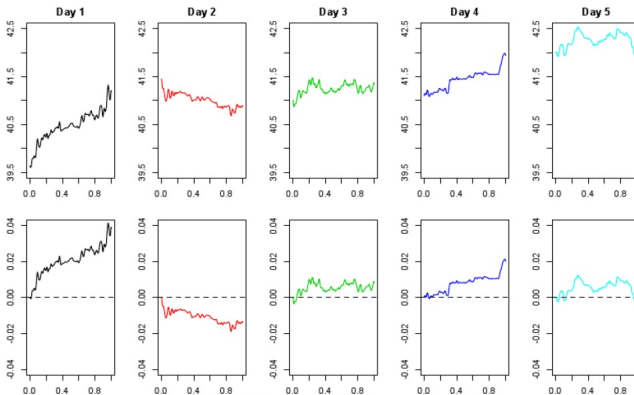
Financial returns II



Log returns for six randomly chosen days of S&P500 trading data. The unit of the time axis is $\Delta = 5\text{min}$ for all figures.

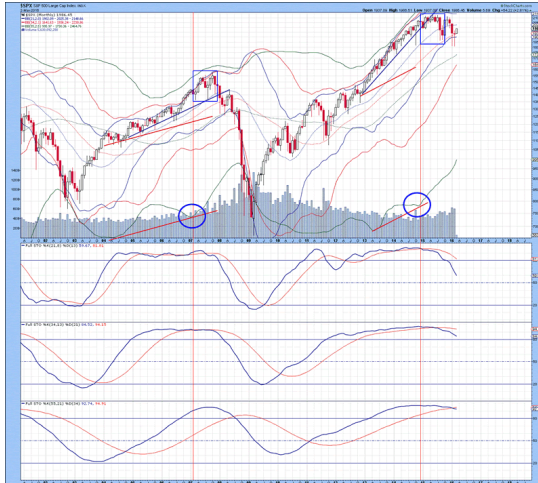
Cumulative Intraday Return (CIDR):

$$r_n(t) = \log[P_n(t)] - \log[P_n(0)].$$



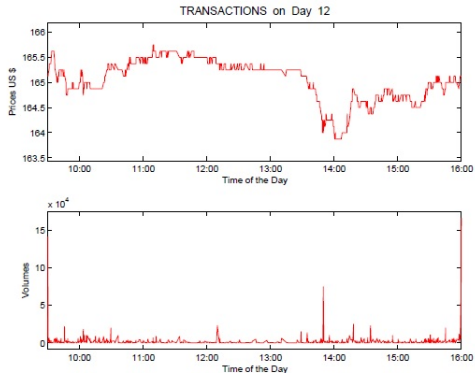
Stock market indexes:

A stock index or stock market index measures the value of a section of the stock market. It is a mathematical construct.



Financial time series of "tick data":

Prices (top) and Volumes (bottom):





Andersson, J. and Newbold, P. (2002).

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